Equivox

Browser control by voice

Speak a browser task. Equivox starts with the first complete command while you continue talking.

A spoken browser session

“Open a new tab, then search for physiotherapy clinics in Edmonton, okay now scroll down.”

1

Open the tab

The first complete clause can execute while speech continues.

2

Run the search

The next clause supplies the query to the new tab.

3

Scroll the results

The next decision uses the page after the search loads.

How Equivox works

01	**Speech**	Apple Speech sends partial transcripts as the user talks. Typed commands enter the same loop.
02	**Page context**	The browser exposes visible controls, field labels, focus, and recent actions.
03	**Decision**	Jev chooses an action and an observed target, then judges whether the command is ready.
04	**Execution**	Python checks the decision. Playwright performs the action and refreshes the context.

What you can ask it to do

Tabs and navigation

“Open a new tab.”
“Go back.”
“Switch to the Gmail tab.”

Forms

“Type Sarah Chen
into the name field.”
“Select Alberta.”

Page interactions

“Search for nearby clinics.”
“Click the second result.”
“Scroll down.”

Page appearance

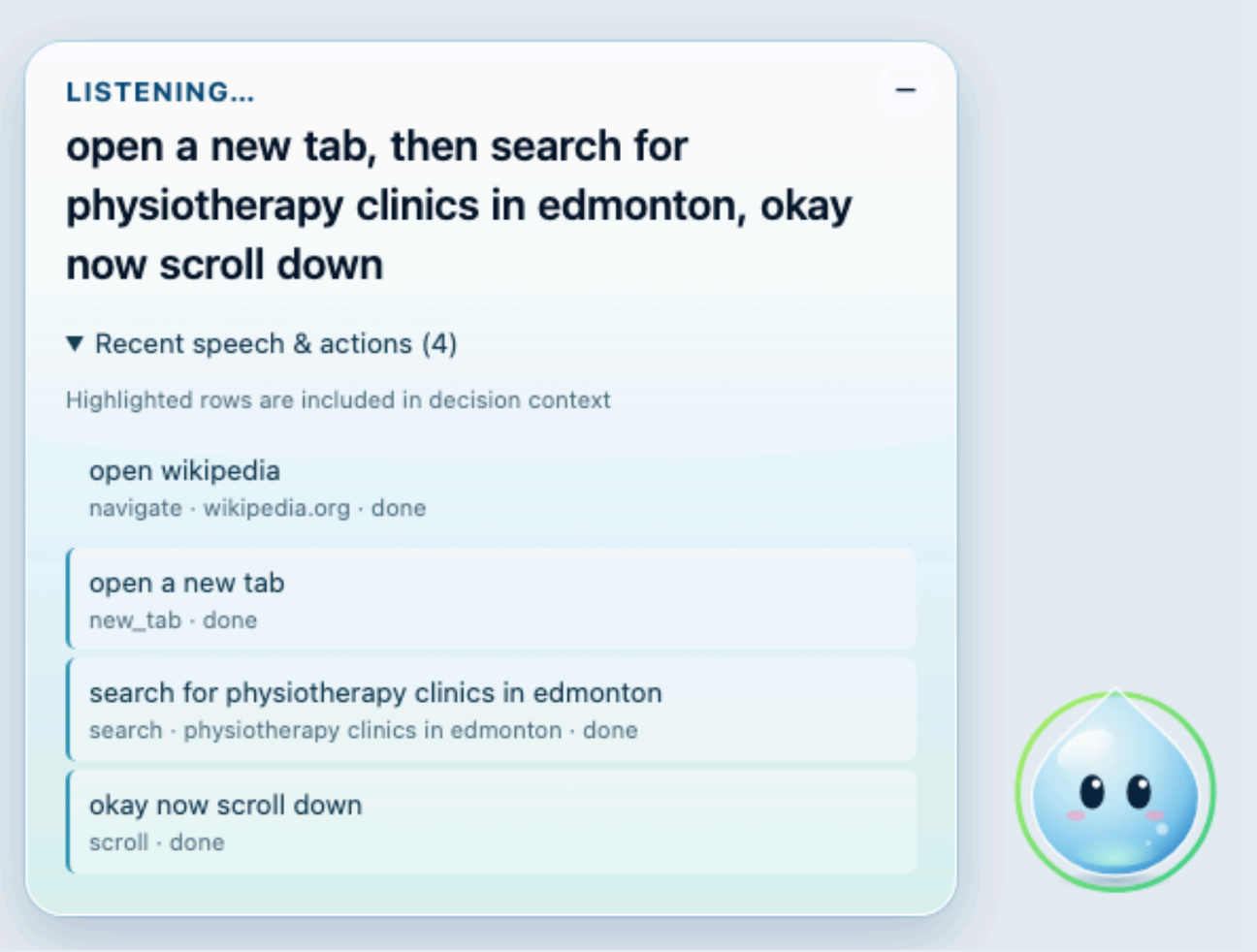
“Make the text bigger.”
“Turn on high contrast.”

Vox keeps the session visible

The floating panel shows the transcript and completed actions.

Highlighted rows show the recent history sent to Jev.

12 actions in decision context
100 actions in session history



Acting during speech

Regular decision checks

An 80 ms scheduling window prevents continuous partials from postponing every decision.

Complete clauses can execute

Equivox can open a tab before the user finishes the rest of the request.

Growing queries wait

Search and typed text wait for a clear boundary, a finished transcript, or enough quiet.

Each action updates the page

The next command uses fresh context. Executed words leave the pending transcript.

What we have measured

187 ms

Median Jev decision time

Seven synthetic requests
177 to 406 ms observed range

294

Passing regression tests

Includes command chains, browser controls, and history across navigation.

These timings cover model decisions only. Full speech-to-action latency still needs measurement with real microphone input.

What still needs work

Real speech measurement	Replay actual recognition updates and measure when each action starts.
Noise and corrections	Test hesitation, changing words, and different microphone conditions.
Complex page controls	Improve custom dropdowns, rich editors, and interfaces drawn on canvas.
Visual grounding	Evaluate the optional screenshot path on controls that DOM text cannot explain.